

Javascript for Page Editing

There are a couple ways to use Javascript in **FRONT END**. And I'm going to cover the more basic one; which is changing what appears on the page when the user **interacts with the page**.

- This is called **DOM editing**, Data Object Model.
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So what we're doing is **GRABBING** an *HTML division* via Javascript code. Then **choosing to do something with it**.

- **Usually**, we change the CSS class, or change the content inside the div.
-

What do I want to do though?

- Let's say I want to **TOGGLE** the *table of contents open and close*.

This would be achieved via, a **NEW** class that simply makes the sidebar **0 width**, *and some other finicky things*.

So let's go to our CSS page. And we make a **NEW** class, **that uses the old class too in it's name**.

```
.table-of-contents.closed {}
```

We have **TWO DOTS**, the first one is the original class, and the 2nd one is the **ADDITION** in the name.

- In the HTML, it'd look like **class="table-of-contents closed"**

So, let's attach the **new attributes** we want to *this class*.

```
.table-of-contents.closed {  
width: 0;  
overflow: hidden;  
transition: width 0.3s ease-in-out;  
}
```

Overflow hidden makes it not go finicky when it opens back up.

Now how do we even toggle it?

- Let's just add a button on the top left of the **content section**

Real quick, add position relative to .content, then paste in this.

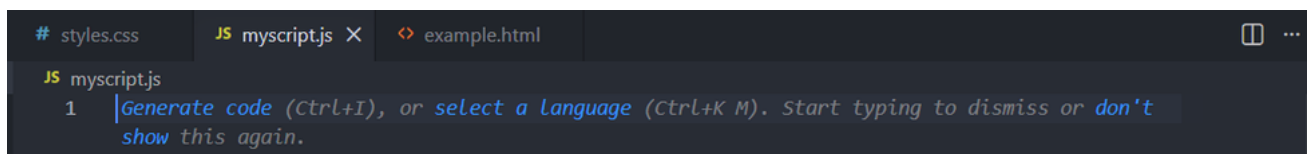
```
.close-btn {  
  position: absolute;  
  top: 16px;  
  left: 16px;  
  color: rgb(221, 216, 216);  
  padding: 10px;  
  background-color: #3d3f3f;  
  border-radius: 20%;  
  font-size: 30px;  
  cursor: pointer;  
}
```

Then, add it as a button in the contents division in the HTML.

```
<button class="close-btn">&gt;</button>
```

NOW, we can get to using Javascript.

- Make a new Javascript file.



First, we grab the TWO elements that are involved in the interaction as variables.

- We're not going to learn the language really, just the snippets that are commonly used.

The elements that are involved are the TABLE OF CONTENTS, and CLOSE BUTTON.

- Grab variables like this: `const element = document.querySelector('.class-name');`

So, like this →

```
const tableOfContents = document.querySelector('.table-of-contents');  
const closeBtn = document.querySelector('.close-btn');
```

Then, we use something called an EVENT listener. Which is an action by the user, here we're passing in the CLICK event.

This is the *format*.

- **thingImClicking**, is our button i.e. what *we're listening to be clicked on*, which we saved as **closeBtn**.

```
thingImClicking.addEventListener('click', () => {  
  
  // One of the BELOW go here.  
  
})
```

INSIDE those {}, we can either do our CLASS change or the change content. But we're gonna do class change.

- We can do ONE of three things; ADD the class, TOGGLE the class, REMOVE the class.

```
element.classList.add('active');  
element.classList.remove('active');  
element.classList.toggle('active');
```

So, we want the TOGGLE; so every time we press the button it removes and adds the class back.

- So it looks like this.

```
const tableOfContents = document.querySelector('.table-of-contents');  
const closeBtn = document.querySelector('.close-btn');  
  
closeBtn.addEventListener('click', () => {  
  tableOfContents.classList.toggle('closed');  
});
```

Now we need the HTML page to be able to access this script; *put this at the bottom of our body*.

```
<script src="myscript.js"></script>
```

But this makes the sidebar vanish, but not have the content section FILL IT UP.



So, we ACTUALLY need to grab the **SKELETON division**, and make it's new class.

- Make a NEW class, inside it have this NEW COLUMN SIZE.

```
.skeleton.closed {  
  grid-template-columns: 0px 1fr;  
}
```

Delete the old `.table-of-contents.closed`

And let's go back to **Javascript**, change the element we're interacting with to **SKELETON**.

- It will now look like this.

```
const closeTableOfContents = document.querySelector('.skeleton');  
const closeBtn = document.querySelector('.close-btn');  
closeBtn.addEventListener('click', () => {  
  closeTableOfContents.classList.toggle('closed');  
});
```

And last issue, it doesn't hide the content when we click it.

- That's why we can use the "tree" classes.

So, you know how when I close the **table of contents**; **skeleton** will have the class "**skeleton closed**"

- We can TELL **table of contents**, that **skeleton** is currently closed and CHANGE the **table of contents** in a new class.

This means, assuming the **PARENT** division has the class **skeleton closed**, the normal skeleton closed class will inherit these properties.

```
.skeleton.closed .table-of-contents {  
  overflow: hidden;  
}
```

Now, It's gone on the side.

Lastly, **let's cover** editing the content.

- And we'll use a **CONDITION**, **because we'd want them to undo things**.

Simply, I want the **ARROW** on the **closeBtn** to flip, **on if it's closed or not**.

In the **SAME** Javascript file, add this new code **STARTS THE SAME**.

```
closeBtn.addEventListener('click', () => {})
```

NOW, instead we do an **IF STATEMENT**.

- **DEPENDING** on **if our skeleton is closed or not**; we use that to **CHANGE** what's inside. (2nd line)

```
closeBtn.addEventListener('click', () => {  
  if (closeTableOfContents.classList.contains('closed')) {  
    closeBtn.innerHTML = '&gt;'; // >  
  } else {  
    closeBtn.innerHTML = '&lt;'; // <  
  }  
});
```

That's the **BASICS** on **FRONT END** development. **If you want to continue focus on this list**.

- Javascript → localStorage
- CSS → Animations

But, for now that's it.